Geographic Software Design
Spring 2025 - Geog 422/510
Department of Geography & Sustainability
University of Tennessee, Knoxville
(3 credit hours)

Course Information

Faculty Contact Information

- Instructor Name: Dr. Qiusheng Wu
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- Email: qwu18@utk.edu
- Phone Number: (865)221-8824
- Website: https://wetlands.io
- Office Hours: Mondays & Wednesdays, 9:00 AM - 10:00 AM, BGB-309

Course Schedule and Resources

- Lectures: MWF, 1:50 PM - 2:40 PM, BGB-404
- Course Website: https://geog-510.gishub.org
- GitHub Repository: https://github.com/giswqs/geog-510
- Q&A Discussion Board: https://github.com/giswqs/geog-510/discussions

Course Description

This course offers a comprehensive introduction to creating and designing open-source Python packages for geospatial analysis and interactive mapping. Students will gain hands-on experience with popular Python development environments, including Visual Studio Code, Jupyter Notebook, and Google Colab, to streamline the development, packaging, and distribution of their code. The course emphasizes leveraging existing geospatial libraries, such as geemap and leafmap, as models for best practices, while fostering contributions to the open-source community.

Throughout the course, students will learn how to design a fully functional Python package, complete with an accompanying documentation website that can be freely hosted on GitHub. The curriculum includes essential topics such as version control with Git, writing clean and reusable code, generating documentation, and sharing projects with collaborators and the broader Python ecosystem.

By the end of the course, students will have developed their own Python package, gaining practical skills highly valued in technical roles such as GIS analyst, data scientist, and software developer. These skills will not only enhance their resumes but also position them competitively in the job market for cutting-edge geospatial and software development roles.

Student Learning Outcomes

Upon successfully completing the course, students will be able to:

- Apply git for effective version control
- Design and develop open-source Python packages for geospatial analysis
- Leverage open-source geospatial libraries
- Host and maintain open-source code on GitHub
- Deploy Python packages to PyPI and conda-forge
- Develop and host documentation websites
- Collaborate on open-source projects
- Contribute meaningfully to the open-source community
- Enhance employability in geospatial and software development roles

How to Be Successful in This Course

Here are some tips to be successful in this course, including but not limited to:

- Attend weekly lectures and lab sessions.
- Start lab assignments early—don't wait until the last minute.
- Practice coding daily to reinforce skills.
- Use the GitHub Discussion board to ask questions and collaborate.
- Visit office hours for help and guidance.

Learner Expectations

- Be prepared for all classes by reviewing materials in advance.
- Treat classmates and instructors with respect.
- Actively participate in class activities and discussions.
- Follow the UT Honor Code in all coursework.

Instructor Expectations

- Come prepared to all classes and labs.
- Provide fair and equitable evaluations.
- Respect and support all students.
- Create meaningful and engaging learning activities.
- Adhere to university codes of conduct.

Course Requirements

Recommended Texts

- Tomas Beuzen & Tiffany Timbers (2022). *Python Packages*. https://py-pkgs.org
- Qiusheng Wu (2024). *Introduction to GIS Programming*. https://geog-312.gishub.org
- Qiusheng Wu (2024). *Geemap*. https://geemap.org
- Qiusheng Wu (2024). *Leafmap*. https://leafmap.org

Software Tools

- Git
- Visual Studio Code
- Jupyter Notebook
- PyCharm

Open-Source Geospatial Packages

- geemap
- leafmap
- whitebox
- ipyleaflet
- folium
- maplibre
- geopandas
- xarray
- rioxarray
- localtileserver

Course Communication Policy

Email Communication Policy

All course-related questions, except those of a personal and private nature, ***must*** be posted at https://github.com/giswqs/geog-510/discussions. Emails of a personal or private nature sent to the instructor should include "GEOG-510" in the email subject line. The instructor reserves ***30 extra credit points*** (3% of the total grade) for those students who actively respond to the

questions from other students. The instructor will strive to respond to questions within 24 hours during the week and 48 hours on the weekend.

Prerequisites and Required Skills

Prerequisites

While not required, completing one or more of the following courses is recommended for better preparation:

- **GEOG-311**: Geovisualization and Geographic Information Science
- **GEOG-312**: Introduction to GIS Programming
- **GEOG-411**: Intermediate GIS

Essential Skills

Students are expected to be computer literate and have a working knowledge of an operating system such as Windows, macOS, or Linux. Additionally, students are responsible for self-learning basic software functions as needed.

Specifically, students should be able to:

- Use email effectively to communicate with the instructor.
- Search the Internet for course-relevant data and information.
- Download files from Canvas and handle file compression tasks (e.g., zip, tar) using software like 7-Zip or WinZip.
- Submit lab assignments on Canvas following the required format.

Grading Scheme

Grading Components

ASSIGNMENT	POINTS	PERCENTAGE
Lab assignments	600	60%
Midterm exam	150	15%
Final project	250	25%
Total	1000	100%

Lab Assignments (60%)

- The course includes **10 lab assignments**, accounting for 60% of the total grade.
- **Late Submissions**: A **20% penalty** will be applied per day for late labs.
- Formatting: All lab assignment write-ups must be typed. Handwritten work will not be accepted.
- **Submission Responsibility**: Students must ensure the correct files are submitted before the due date.
- **Note:** No additional files will be accepted after an assignment has been graded.
- **Grade Disputes**:
- Students must notify the instructor within **one week** of the grading date if they wish to contest a grade.
- Requests made more than one week after the grading date will **not** be considered, regardless of the reason.

Exams (15%)

- The course includes **one midterm exam** worth 15% of the total grade.
- **Exam Policies**:
- The exam must be taken on the designated day unless prior arrangements are made.
- Unexcused absences on the exam day will result in a grade of **zero**.

Final Project (25%)

- Students will independently design a project of their choosing.

- **Deliverables**:
- A final project presentation will be required.
- The project will be evaluated on creativity, technical implementation, and presentation quality.
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Extra Credit (3%)

- **Participation Incentive**: Students can earn up to 30 extra credit points (3% of the total grade) for actively responding to course-related questions from peers.
- Questions should be posted at the GitHub Discussion Board.
- Personal or private queries should be addressed directly to the instructor.

Grading Scale

LETTER GRADE	PERCENTAGE
A	94.00 – 100%
A-	90.00 – 93.99%
B+	87.00 – 89.99%
B	84.00 – 86.99%
B-	80.00 – 83.99%
C+	77.00 – 79.99%
C	74.00 – 76.99%
C-	70.00 – 73.99%
D+	67.00 – 69.99%
D	64.00 – 66.99%
D-	61.00 – 63.99%
F	0.00 – 60.99%

Expectations for Undergraduate and Graduate Students

In this course, both undergraduate and graduate students will engage with the same core curriculum and assignments. However, the final project submitted by graduate students will be held to a higher standard, reflecting the advanced level of their studies. Graduate students are expected to demonstrate a deeper understanding of geospatial principles, more comprehensive analysis, and greater complexity in their Python package design and implementation.

Policies on Make-up Exams

Exams may not be made up except under valid university excuses such as:

- **Medical** – Documentation must be provided on official letterhead, signed and dated by a licensed medical or mental health provider, including the dates of treatment and student's anticipated return date. Medical records should not be provided and will not be accepted as documentation.
- **Death in the Family** – Documentation should include an obituary, memorial service pamphlet, or other items that identify the student as related to the deceased and include the date of expiration, date of funeral/memorial service, and the city and state where the service was held.
- **Military/Legal Obligations** – Only military orders or court papers will be accepted. The orders/papers must be signed and dated by a commanding officer/designee/judge/lawyer and include dates of departure and return.

- **For attending academic conferences** – The student must inform the instructor two weeks before the exam or quiz and submit a letter from a faculty member stating the importance of the conference.

Please note: The instructor reserves the right to revise, alter or amend this syllabus as necessary. Students will be notified in writing/email of any such changes.

University Policies

Academic Integrity

Each student is responsible for their personal integrity in academic life and for adhering to UT's Honor Statement. The Honor Statement reads: "An essential feature of the University of Tennessee, Knoxville is a commitment to maintaining an atmosphere of intellectual integrity and academic honesty. As a student of the university, I pledge that I will neither knowingly give nor receive any inappropriate assistance in academic work, thus affirming my own personal commitment to honor and integrity."

Your Role In Improving The Course Through Assessment

At UT, it is our collective responsibility to improve the state of teaching and learning. During the semester you may be requested to assess aspects of this course, either during class or at the completion of the class, and through the TNVoice course evaluation. Please take the few moments needed to respond to these requests as they are used by instructors, department heads, deans and others to improve the quality of your UT learning experience.

Students With Disabilities

The University of Tennessee, Knoxville, is committed to providing an inclusive learning environment for all students. If you anticipate or experience a barrier in this course due to a chronic health condition, a learning, hearing, neurological, mental health, vision, physical, or other kinds of disability, or a temporary injury, you are encouraged to contact Student Disability Services (SDS) at 865-974-6087 or sds@utk.edu. An SDS Coordinator will meet with you to develop a plan to ensure you have equitable access to this course. If you are already registered with SDS, please contact your instructor to discuss implementing accommodations included in your course access letter.

Accessibility Policy And Training

The University of Tennessee, Knoxville, provides reasonable accommodations for individual students with disabilities through its office of Student Disability Services. The university is also committed to making information and materials accessible, when possible. Resources and assistance to support these efforts can be found at http://accessibility.utk.edu/.

Wellness

The Center for Health Education and Wellness empowers all Volunteers to thrive by cultivating personal and community well-being. The Center can answer questions about general wellness, substance use, sexual health, healthy relationships, and sexual assault prevention. The Student Counseling Center is the university's primary facility for personal counseling, psychotherapy, and psychological outreach and consultation services.

Any student who has difficulty affording hygiene products, groceries, or accessing sufficient food to eat every day is urged to contact the Big Orange Pantry for support. The Big Orange Pantry, located in Greve Hall, is a free resource for all students, faculty, and staff, no matter how great or small their need is. Students who need emergency financial assistance can also request funding from the Student Emergency Fund.

Students who are experiencing non-academic difficulty or distress and need assistance should call 974-HELP or submit an online referral. The 974-HELP team specializes in aligning resources and support for students experiencing mental health distress.

Emergency Alert System

The University of Tennessee is committed to providing a safe environment to learn and work. When you are alerted to an emergency, please take appropriate action. Learn more about what to do in an emergency and sign up for UT Alerts. Check the emergency posters near exits and elevators for building-specific information. In the event of an emergency, the course schedule and assignments may be subject to change. If changes to graded activities are required, reasonable adjustments will be made, and you will be responsible for meeting revised deadlines.

COVID-19 Guidelines

CDC guidance recognizes the changing dynamics of living in a world with COVID-19. It rates COVID-19 community levels as low, medium, and high, with recommendations at each level about the use of masks and other precautions. At all levels of community spread, staying up-to-date with vaccination is the best way to protect yourself from serious illness and to limit the spread of COVID-19. Wearing a mask is always an option for any individual who chooses to do so, and the CDC recommends that those with a high risk of severe illness talk with their health care providers. If you are sick, please stay in, avoid being around others as much as possible, and contact your health care provider for any symptoms that are worsening, moderate to severe, or concerning to you. For more information about vaccination or to self-report an illness and receive support, visit https://studenthealth.utk.edu/CommunityHealth. For more information about COVID-19, visit https://studenthealth.utk.edu/covid-19.